

Jiaying Lu

Assistant Professor
Center for Data Science
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Google Scholar

RESEARCH INTERESTS

My research interests span AI for healthcare, knowledge graph & LLM co-learning, multimodal learning, and data mining. I have authored 30+ peer-reviewed publications in leading AI/ML and biomedical/health informatics venues, receiving over 1,000 citations with an h-index of 12.

EDUCATION BACKGROUNDS

- Ph.D. [Computer Science & Informatics](#), Emory University, USA, 2019–2024
Dissertation: Learning Structured Knowledge from Real-World Data without Excessive Annotations
Advisor: [Dr. Carl Yang](#)
- M.Eng. Electronics & Communication Engineering, Beijing University of Posts and Telecommunications, China, 2014–2017
Exchange student at Aalto University, Finland, Sep 2015–Jan 2016
- B.Sc. Information Engineering, Beijing University of Posts and Telecommunications, China, 2010–2014

ACADEMIC APPOINTMENTS

- 2024– [Center for Data Science](#), Nell Hodgson Woodruff School of Nursing, Emory University.
Research-Track Assistant Professor

SELECTED HONORS & AWARDS

- 2024– Emory Undergraduate Conference Presentation Grants (Cr. Katie Shao, Victor Li). *Emory College of Arts and Sciences Undergraduate Research Programs*.
- 2024 **Young Scientist Excellence Award** (4th-place winner). *MidSouth Computational Biology and Bioinformatics Society*.
- 2022 NSF Registration Award. *31st ACM International Conference on Information and Knowledge Management*.
- 2022 NSF Student Travel Award. *28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*.
- 2022 Registration Award. *Georgia Statistics Day 2022*.
- 2021–24 Professional Development Support Funds. *Laney Graduate School at Emory University*.

SELECTED PUBLICATIONS

- (a) **Jiaying Lu**, Xiangjue Dong, and Carl Yang. Weakly supervised concept map generation through task-guided graph translation. *IEEE Transactions on Knowledge and Data Engineering*, 35(10): 10871–10883, March 2023. doi: 10.1109/TKDE.2023.3252588 (IF=8.9, *CCF-A*, *CAS Q1-top*)
- (b) **Jiaying Lu**, John R. Bowblis, Shuang Li, Yong-Fang Kuo, Jennifer Heston-Mullins, James S. Goodwin, and Huiwen Xu. Patterns and predictors of reopening nursing homes to visitors during the covid-19 pandemic. *Journal of the American Medical Directors Association*, 2025a. doi: 10.1016/j.jamda.2025.105918 (IF=3.8, *CAS Q2*)

- (c) **Jiaying Lu**, Jiaming Shen, Bo Xiong, Wenjing Ma, Steffen Staab, and Carl Yang. Hiprompt: Few-shot biomedical knowledge fusion via hierarchy-oriented prompting. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR'23 (Short Paper), page 2052–2056, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9781450394086. doi: 10.1145/3539618.3591997 (h-index=103, CCF-A)
- (d) Shuyue Jiang, Wenjing Ma, Shaojun Yu, Chang Su, Runze Yan, and **Jiaying Lu**[✉]. Integrating epigenetic and phenotypic features for biological age estimation in cancer patients via multimodal learning. In *Proceedings of The 19th IEEE International Conference on Bioinformatics and Biomedicine*, (BIBM'25), 2025. doi: 10.1109/bibm66473.2025.11356564 (h-index=31, CCF-B)
- (e) Rongjia Zhou, Chengzhuo Li, Carl Yang, and **Jiaying Lu**[✉]. Clinnoteagents: An llm multi-agent system for predicting and interpreting heart failure 30-day readmission from clinical notes. In *Proceedings of The AMIA 2026 Amplify Informatics Conference*, (AMIA'26), 2026 (h-index=30, CCF-C)

GRANTS

Ongoing Research Support

- 2026 *Leveraging Large Language Models to Mine Environmental Exposures and Pediatric Health Outcomes in Georgia Child Care Settings* (Lu)
Pilot Grant (\$12,000)
NIEHS's Center for Children's Health Assessment, Research Translation, and Combating Environmental Risk
Role: Principal Investigator
- 2024–27 *Incorporating Structured Knowledge in Large Language Models for Precision Healthcare* (Lu)
Startup Grant (\$100,000)
Emory Nursing Data Science Foundation
Role: Principal Investigator

Recently Completed Projects

- 2025 *Understanding Generative AI's Impact to Users from Various Religious Background* (Lu)
Seed Grant (\$5,000)
Atlanta Interdisciplinary Artificial Intelligence Network
Role: Principal Investigator

PUBLICATIONS

Equal contribution denoted with *, corresponding authors with ✉.

Journal Articles

- (J7) **Jiaying Lu**, John R. Bowblis, Shuang Li, Yong-Fang Kuo, Jennifer Heston-Mullins, James S. Goodwin, and Huiwen Xu. Patterns and predictors of reopening nursing homes to visitors during the covid-19 pandemic. *Journal of the American Medical Directors Association*, 2025a. doi: 10.1016/j.jamda.2025.105918 (IF=3.8, CAS Q2)
- (J6) Runze Yan, Hanqi Luo, **Jiaying Lu**, Darren Liu, Terryl J Hartman, and Xiao Hu. Dietai24 as a framework for comprehensive nutrition estimation using multimodal large language models. *Communications Medicine*, 2025a. doi: 10.1038/s43856-025-01159-0 (IF=5.4, CAS Q2)
- (J5) Hejie Cui*, **Jiaying Lu***, Ran Xu*, Shiyu Wang, Wenjing Ma, Yue Yu, Shaojun Yu, Xuan Kan, Chen Ling, Liang Zhao, Zhaohui S. Qin, Joyce Ho, Tianfan Fu, Jing Ma, Mengdi Huai, Fei Wang, and Carl Yang. A survey on knowledge graphs for healthcare: Resources, applications, and promises. *Journal of Biomedical Informatics*, July 2025. doi: 10.1016/j.jbi.2025.104861 (A short version has been presented at the ICML 2023 Workshop on Interpretable Machine Learning in Healthcare. IF=4.0, CAS Q2)

- (J4) Chen Ling, Xujiang Zhao, **Jiaying Lu**, Chengyuan Deng, Can Zheng, Junxiang Wang, Tanmoy Chowdhury, Yun Li, Hejie Cui, Xuchao Zhang, Tianjiao Zhao, Amit Panalkar, Dhagash Mehta, Stefano Pasquali, Wei Cheng, Haoyu Wang, Yanchi Liu, Zhengzhang Chen, Haifeng Chen, Chris White, Quanquan Gu, Jian Pei, Carl Yang, and Liang Zhao. Domain specialization as the key to make large language models disruptive: A comprehensive survey. *ACM Computing Surveys*, 2025. doi: 10.1145/3764579 (Cited by 2024 Economic Report of the President of USA. IF=23.8, CAS Q1-top)
- (J3) **Jiaying Lu**, Ran Xiao, Xiao Hu, and Duc H. Do. Artificial intelligence in cardiac telemetry. *BMJ Heart*, 2025b. doi: 10.1136/heartjnl-2024-323947 (IF=5.1, CAS Q2-top)
- (J2) **Jiaying Lu**, Xiangjue Dong, and Carl Yang. Weakly supervised concept map generation through task-guided graph translation. *IEEE Transactions on Knowledge and Data Engineering*, 35(10): 10871–10883, March 2023. doi: 10.1109/TKDE.2023.3252588 (IF=8.9, CCF-A, CAS Q1-top)
- (J1) Wenjing Ma, **Jiaying Lu**, and Hao Wu. Cellcano: supervised cell type identification for single cell atac-seq data. *Nature Communications*, 14(1):1864, April 2023. ISSN 2041-1723. doi: 10.1038/s41467-023-37439-3 (2023 ASA SSGG Distinguished Student Paper Award. IF=14.7, CAS Q1-top)

Conference Papers

- (C26) Shuting Jin, Xu Guo, Anqi Huang, Hongxin Xiang, Yajie Meng and Junlin Xu[✉], **Jiaying Lu[✉]**, and Xiaobo Sun[✉]. Diffpro: Decoupled generative prior with diffusion models for efficient few-shot drug synergy prediction. In *Proceedings of 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, (KDD'26), 2026 (h-index=124, CCF-A)
- (C25) Yuzhang Xie, Keqi Han, Yunpeng Xiao, Hejie Cui, Guanchen Wu, Ziyang Zhang, Kai Shu, **Jiaying Lu**, Xiao Hu, and Carl Yang. Ehrbench: An automated and reliable ehr-based benchmark for clinical decision making with llms. In *Proceedings of 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, (KDD'26), 2026 (h-index=124, CCF-A)
- (C24) Mingchen Shao, Yuzhang Xie, Carl Yang, and **Jiaying Lu[✉]**. Llm-mine: Large language model based alzheimer's disease and related dementias phenotypes mining from clinical notes. In *Proceedings of The AMIA 2026 Annual Symposium*, (AMIA'26), 2026 (h-index=30, CCF-C)
- (C23) Xinghao Li, Jiacong Mi, **Jiaying Lu[✉]**, and Hao Wu[✉]. Modeling heterogeneous glycemic responses to therapy from multimodal cgm events in type 2 diabetes via a transformer-vae. In *Proceedings of The AMIA 2026 Annual Symposium*, (AMIA'26), 2026b (h-index=30, CCF-C)
- (C22) Rongjia Zhou, Chengzhuo Li, Carl Yang, and **Jiaying Lu[✉]**. Clinnoteagents: An llm multi-agent system for predicting and interpreting heart failure 30-day readmission from clinical notes. In *Proceedings of The AMIA 2026 Amplify Informatics Conference*, (AMIA'26), 2026 (h-index=30, CCF-C)
- (C21) Ruiyu Wang, Tuan Vinh, Ran Xu, Yuyin Zhou, **Jiaying Lu**, Francisco Pasquel, Mohammed Ali, and Carl Yang. Knowledge graph augmented large language models for disease prediction. In *Proceedings of The AMIA 2026 Amplify Informatics Conference*, (AMIA'26), 2026 (h-index=30, CCF-C)
- (C20) Shuyue Jiang, Wenjing Ma, Shaojun Yu, Chang Su, Runze Yan, and **Jiaying Lu[✉]**. Integrating epigenetic and phenotypic features for biological age estimation in cancer patients via multimodal learning. In *Proceedings of The 19th IEEE International Conference on Bioinformatics and Biomedicine*, (BIBM'25), 2025. doi: 10.1109/bibm66473.2025.11356564 (h-index=31, CCF-B)
- (C19) Wenjing Ma, Xiaoqing Yu, **Jiaying Lu**, Jing Zhang, Xiang Zhou, and Xuefeng Wang. scpankd: Distilling pan-cancer knowledge for enhanced t cell subtypes annotation in single-cell transcriptomics data. In *Proceedings of The 19th IEEE International Conference on Bioinformatics and Biomedicine*, (BIBM'25), 2025b. doi: 10.1109/bibm66473.2025.11356354 (h-index=31, CCF-B)
- (C18) Yuzhang Xie, Xu Han, Ran Xu, Xiao Hu, **Jiaying Lu**, and Carl Yang. Hypkg: Hypergraph-based knowledge graph contextualization for precision healthcare. In *Proceedings of The 24th International*

- Semantic Web Conference*, (ISWC'25), pages 328–348, October 2025b. doi: 10.1007/978-3-032-09527-5_18 (h-index=37, CCF-B)
- (C17) Victor Li, Runze Yan, Alex Fedorov, and **Jiaying Lu**[✉]. Ensemble learning with early fusion of kernel-transformed and classical electrocardiogram features for chagas disease detection. In *Proceedings of The 52th International Computing in Cardiology Conference*, (CinC'25), 2025 (**Student Supported by Emory Undergraduate Research Conference Presentation Grant**). h-index=18)
- (C16) Yuhao Xu, Xiaoda Wang, **Jiaying Lu**, Sirui Ding, Defu Cao, Huaxiu Yao, Yan Liu, Xiao Hu, and Carl Yang. Enecg: Efficient ensemble learning for electrocardiogram multi-task foundation model. In *Proceedings of American Medical Informatics Association 2025 Annual Symposium*, (AMIA'25), 2025b (h-index=30, CCF-C)
- (C15) Yuzhang Xie, Hejie Cui, Ziyang Zhang, **Jiaying Lu**, Kai Shu, Fadi Nahab, Xiao Hu, and Carl Yang. Kerap: A knowledge-enhanced reasoning approach for accurate zero-shot diagnosis prediction using multi-agent llms. In *Proceedings of American Medical Informatics Association 2025 Annual Symposium*, (AMIA'25), November 2025a (h-index=30, CCF-C)
- (C14) **Jiaying Lu**[✉], Stephanie R. Brown*, Songyuan Liu, Shifan Zhao, Kejun Dong, Del Bold, Michael Fundora, Alaa Aljiffry, Alex Fedorov, Jocelyn Grunwell, and Xiao Hu. Early risk prediction of pediatric cardiac arrest from electronic health records via multimodal fused transformer. In *47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, (EMBC'25), 2025. doi: 10.1109/EMBC58623.2025.11251673 (h-index=48)
- (C13) Saurabh Kataria, Ran Xiao, Timothy Ruchti, Matthew T Clark, **Jiaying Lu**, Randall Lee, Jocelyn Grunwell, and Xiao Hu. Continuous cardiac arrest prediction in icu using ppg foundation model. In *47th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, (EMBC'25), 2025b. doi: 10.1109/EMBC58623.2025.11254649 (h-index=48)
- (C12) Mingchen Shao, Youjeong Kang, Xiao Hu, Hyunjung Gloria Kwak, Carl Yang, and **Jiaying Lu**[✉]. Mining social determinants of health for heart failure patient 30-day readmission via large language models. In *The 20th World Congress on Medical and Health Informatics - Poster*, (MedInfo'25), 2025 (**Student Supported by Emory Undergraduate Research Conference Presentation Grant**)
- (C11) Yuhao Xu, **Jiaying Lu**, Sirui Ding, Xiao Hu, and Carl Yang. An electrocardiogram multi-task benchmark with comprehensive evaluations and insightful findings. In *The 20th World Congress on Medical and Health Informatics*, (MedInfo'25), 2025a
- (C10) Guanchen Wu, Linzhi Zheng, Han Xie, Zhen Xiang, **Jiaying Lu**, Darren Liu, Delgersuren Bold, Bo Li, Xiao Hu, and Carl Yang. Large language model empowered privacy-protected framework for phi annotation in clinical notes. In *The 20th World Congress on Medical and Health Informatics*, (MedInfo'25), 2025
- (C9) Bo Xiong, **Jiaying Lu**, Yuqicheng Zhu, and Carl Yang. Mash: Maximal separating poincaré hyperplanes for hierarchical imbalanced learning. In *Companion Proceedings of the ACM on Web Conference 2025 (Short Paper)*, (WWW'25), page 1431–1435, New York, NY, USA, 2025. Association for Computing Machinery. ISBN 9798400713316. doi: 10.1145/3701716.3715505 (h-index=112, CCF-A)
- (C8) Yuzhang Xie, **Jiaying Lu**, Joyce Ho, Fadi Nahab, Xiao Hu, and Carl Yang. Promptlink: Leveraging large language models for cross-source biomedical concept linking. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR'24, page 2589–2593, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400704314. doi: 10.1145/3626772.3657904 (h-index=103, CCF-A)
- (C7) **Jiaying Lu**, Shifan Zhao, Wenjing Ma, Hui Shao, Xiao Hu, Yuanzhe Xi, and Carl Yang. Uncertainty-aware pre-trained foundation models for patient risk prediction via gaussian process. In *Companion Proceedings of the ACM Web Conference 2024*, WWW'24, page 1162–1165, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400701726. doi: 10.1145/3589335.3651456 (A short version is selected for the **Young Scientist Excellence Award** at

MCBIOS'24. h-index=112, CCF-A)

- (C6) **Jiaying Lu***, Yongchen Qian*, Shifan Zhao, Yuanzhe Xi, and Carl Yang. Mug: A multimodal classification benchmark on game data with tabular, textual, and visual fields. In *Findings of the Association for Computational Linguistics: EMNLP 2023*, (Findings-EMNLP'23), December 2023. doi: 10.18653/v1/2023.findings-emnlp.354 (h-index=193)
- (C5) **Jiaying Lu**, Jiaming Shen, Bo Xiong, Wenjing Ma, Steffen Staab, and Carl Yang. Hiprompt: Few-shot biomedical knowledge fusion via hierarchy-oriented prompting. In *Proceedings of the 46th International ACM SIGIR Conference on Research and Development in Information Retrieval*, SIGIR'23 (Short Paper), page 2052–2056, New York, NY, USA, 2023. Association for Computing Machinery. ISBN 9781450394086. doi: 10.1145/3539618.3591997 (h-index=103, CCF-A)
- (C4) Xiangjue Dong, **Jiaying Lu**, Jianling Wang, and James Caverlee. Closed-book question generation via contrastive learning. In *17th Conference of the European Chapter of the Association for Computational Linguistics*, (EACL'23), May 2023. doi: 10.18653/v1/2023.eacl-main.230 (h-index=56)
- (C3) **Jiaying Lu** and Carl Yang. Open-world taxonomy and knowledge graph co-learning. In *4th Conference on Automated Knowledge Base Construction*, (AKBC'22), November 2022
- (C2) Hejie Cui, **Jiaying Lu**, Yao Ge, and Carl Yang. How can graph neural networks help document retrieval: A case study on cord19 with concept map generation. In *44th European Conference on Information Retrieval - Short Paper*, (ECIR'22), April 2022. doi: 10.1007/978-3-030-99739-7_9 (h-index=42, CCF-C)
- (C1) **Jiaying Lu** and Jinho D. Choi. Evaluation of Unsupervised Entity and Event Salience Estimation. In *Proceedings of the 34rd International Florida Artificial Intelligence Research Society Conference*, (FLAIRS'21), May 2021. doi: 10.32473/flairs.v34i1.128482

Workshop Papers

- (w6) Victoria Yan, Honor Chotkowski, Fengran Wang, Xinhui Li, Carl Yang, **Jiaying Lu**, Runze Yan, Xiao Hu, and Alex Fedorov. Towards synthesizing normative data for cognitive assessments using generative multimodal large language models. In *NeurIPS 2025 Workshop on GenAI for Health, Potential, Trust, and Policy Compliance*, (GenAI4Health @NeurIPS 2025), December 2025b
- (W5) Songyuan Liu, Ziyang Zhang, Runze Yan, Wei Wu, Carl Yang, and **Jiaying Lu**[✉]. Measuring spiritual values and bias of large language models. In *KDD 2025 Workshop on Large Language Models for Scientific and Societal Advances*, (KDD'25-SciSocLLM), August 2025
- (W4) Zhaoliang Chen, Cheng Ding, Nirbhay Modhe, **Jiaying Lu**, Carl Yang, and Xiao Hu. Adapting a generative pretrained transformer achieves sota performance in assessing diverse physiological functions using only photoplethysmography signals: A gpt-ppg approach. In *AAAI 2024 Spring Symposium on Clinical Foundation Models*, February 2024
- (W3) **Jiaying Lu***, Jinmeng Rao*, Kezhen Chen, Xiaoyuan Guo, Yawen Zhang, Baochen Sun, Carl Yang, and Jie Yang. Evaluation and enhancement of semantic grounding in large vision-language models. In *AAAI 2024 Workshop on Responsible Language Models*, (ReLM@AAAI'24), February 2024 (**Spotlight**)
- (W2) Zhexiong Liu, Jing Zhang, **Jiaying Lu**, Wenjing Ma, and Joyce Ho. Logicprpbank: A corpus for logical implication and equivalence. In *AAAI 2024 Workshop on AI for Education*, (AAAI'24-AI4ED), February 2024b
- (W1) **Jiaying Lu**, Xin Ye, Yi Ren, and Yezhou Yang. Good, better, best: Textual distractors generation for multi-choice VQA via policy gradient. In *CVPR 2022 Workshop on Open-Domain Retrieval Under a Multi-Modal Setting*, (O-DRUM@CVPR'22), June 2022. doi: 10.1109/CVPRW56347.2022.00539

Preprint Reports

- (P10) Victor Li, Yuzhang Xie, Qingyang Zhu, Wenjing Ma, Xiao Hu, Carl Yang, Jinbing Bai, Huiwen Xu, and **Jiaying Lu**[✉]. Ccq: An llm-curated child care quality dataset to support ai research for children's

health, 2026a

- (P9) Saurabh Kataria, Yi Wu, Zhaoliang Chen, Hyunjung Gloria Kwak, Yuhao Xu, Lovely Yeswanth Panchumarthi, Ran Xiao, **Jiaying Lu**, Ayca Ermis, Anni Zhao, et al. Generalist vs specialist time series foundation models: Investigating potential emergent behaviors in assessing human health using ppg signals, 2025a
- (P8) Ruiyu Wang, Yuzhang Xie, Xiao Hu, Carl Yang, and **Jiaying Lu**[✉]. Biomedjimpact: A comprehensive dataset and llm pipeline for ai engagement and scientific impact analysis of biomedical journals, 2025
- (P7) Wenjing Ma, Siyu Hou, Lulu Shang, **Jiaying Lu**, and Xiang Zhou. Optimal transport modeling uncovers spatial domain dynamics in spatiotemporal transcriptomics studies, 2025a
- (P6) Ximing Ran, Jie Xu, Peng Jin, Zhaohui Qin, Zhexing Wen[✉], and **Jiaying Lu**[✉]. Llmbda: Large language model for biological domains annotation of gene ontology, 2025
- (P5) Runze Yan*, Guanlin Dai*, Yufen Lin, Yuhua Wu, **Jiaying Lu**, Xiao Hu, and Canhua Xiao. Predict epigenetic aging at an early stage: A case study to predict post-radiotherapy aging in head and neck cancer patients, 2025
- (P4) Shifan Zhao, Chen Yue, Jiexin Zheng, and **Jiaying Lu**[✉]. Privacy-preserving synthetic data generation with large language models for enhanced discourse-based adrd assessment, 2025
- (P3) Shifan Zhao, **Lu, Jiaying**, Carl Yang, Edmond Chow, and Yuanzhe Xi. Efficient two-stage gaussian process regression via automatic kernel search and subsampling. *arXiv:2405.13785*, 2024
- (P2) Darren Liu, Cheng Ding, Delgersuren Bold, Monique Bouvier, **Jiaying Lu**, Benjamin Shickel, Craig S. Jabaley, Wenhui Zhang, Soojin Park, Michael J. Young, Mark S. Wainwright, Gilles Clermont, Parisa Rashidi, Eric S. Rosenthal, Laurie Dimisko, Ran Xiao, Joo Heung Yoon, Carl Yang, and Xiao Hu. Evaluation of general large language models in contextually assessing semantic concepts extracted from adult critical care electronic health record notes. *arXiv:2401.13588*, 2024a
- (P1) Guangji Bai, Zheng Chai, Chen Ling, Shiyu Wang, **Jiaying Lu**, Nan Zhang, Tingwei Shi, Ziyang Yu, Mengdan Zhu, Yifei Zhang, Carl Yang, Yue Cheng, and Liang Zhao. Beyond efficiency: A systematic survey of resource-efficient large language models. *arXiv:2401.00625*, 2024

Patents

- 2023 Jinmeng Rao, **Jiaying Lu**, Kezhen Chen, Jie Yang, Yawen Zhang, Xiaoyuan Guo. “A Simulation-Feedback Framework to Mitigate Hallucinations in Multimodal Generative Models”. *Filed 2023. X-52882-00-US*.

Book Chapters

- 2026 **Jiaying Lu**. “Chapter 5. Large Language Models for Clinical Text.” In *Generative AI for Healthcare: From Foundation Models to Clinical Integration* by Carl Yang, Daniel Ting, and Shawn Murphy.

INVITED TALKS

- 2025 “Leveraging Large Language Models to Extract Clinical Knowledge from EHR Notes for Health Outcomes Analysis”. **Annual Conference**. CHAPMS. Virtual Event. Aug 9, 2025.
- 2024 “Leveraging Language Models for Predicting Cardiac Arrest with Multimodal Electronic Health Records in Pediatric ICU”. **Research Seminar**. Children’s Healthcare of Atlanta. Atlanta, GA, USA. Aug 18, 2024.
- 2023 “Retrieval Augmented Generation for Biomedical Entity Alignment via Large Language Models”. **Doctoral Symposium**. 2023 Conference on Health, Inference, and Learning. Boston, MA, USA, Jun 24, 2023.
- 2022 “Learning to Construct Structured Knowledge”. **Data Science Forum**. School of Data Science, The Chinese University of Hong Kong. Palo Alto, CA, USA. Dec 17, 2022.

TEACHING EXPERIENCES

Courses Taught

- 2025 Co-Instructor. *NRSG 745 Information and Healthcare Technologies*. Emory University.
- 2025 Co-Instructor. *NRSG 753 Advanced Computation for Nursing Research I*. Emory University.
- 2021 Teaching Assistant. *CS570 Data Mining*. Emory University.
- 2020 Teaching Assistant. *CS253 Data Structure and Algorithms*. Emory University.
- 2020 Teaching Assistant. *CS170 Introduction to Computer Science I*. Emory University.

Graduate Student Advisement

- 2025– Ximing Ran, Ph.D. in Computer Science& Informatics, Emory University (Research Advisor)
- LLM Enhanced Biological Interpretation for Single Cell Gene Expression Data.
- 2025–26 Siyu Zhai, M.S. in Biostatistics& Bioinformatics, Emory University (*RSPH-APE Research Advisor*).
- Child Care Quality Identification via Ordinal Logistic Regression.
- 2024–25 Andrew Lu, Ph.D. in Nursing AI, Emory University (*Qualifying Exam Committee Member*).
- Early Detection of Occlusion Myocardial Infarction through Responsible AI.
- 2024–25 Shuyue Jiang, M.S. in Data Science, Emory University (*Research Advisor*).
- Chronological Age Prediction based on DNA Methylation and Clinical Features.

Undergraduate Advisement

- 2024– Victor Li, B.S. in Mathematics/Computer Science, Emory University (*Research Advisor*).
- Data-centric AI Methods to Enhance ECG-based Chagas Disease Detection.
- 2025– Jason Li, B.S. in Data and Decision Sciences, Emory University (*Honor Thesis Committee Co-Chair*).
- Emory Undergraduate Research Pathway Funding. 2025 Summer.
 - Chronological Age Prediction of Cancer Patients based on Epigenomics and Transcriptomics.
- 2024– Jasmine Zhou, B.S. in Data and Decision Sciences, Emory University (*Research Advisor*).
- Multi-Agent Clinical Note Analytics.
- 2024–25 Jonathan Wang, B.S. in Computer Science, Emory University (*Honor Thesis Committee Member*).
- Healthcare Journal Impact Factor Trend Prediction: Will Journals Embracing Artificial Intelligence Gain More Impacts?
- 2024–25 Simon Liu, B.S. in Computer Science, Emory University (*Honor Thesis Committee Member*).
- Representing Human Contact as Hypergraph: A Deep Learning Approach.
- 2024–25 Tung Dinh, B.S. in Computer Science, Emory University (*Honor Thesis Committee Member*).
- 2024–25 Mingchen Shao, B.S. in Computer Science, Emory University (*Research Advisor*).
- Large Language Model Enhanced Automated Phenotype Extraction from Clinical Notes.

PROFESSIONAL EXPERIENCES

- 2023 **AI Residency.** Mountain View, CA, USA. May–Sep 2023.
Mineral Research Team, **Google X**.
Supervisors: [Dr. Jinmeng Rao](#), [Dr. Kezhen Chen](#).
- Data-centric approach for mitigating hallucination in multimodal large language models.
- 2022 **Applied Scientist Intern.** Sunnyvale, CA, USA. May–Aug 2022.
Auto Machine Learning Team, **Amazon.com**.
Supervisors: [Dr. Andrew Borthwick](#), [Nick Erickson](#).
- Multi-modal ensemble models for large-scale product classification problems.
- 2018–19 **Visiting Scholar.** Phoenix, AZ, USA. Aug 2018–Jan 2019.
[Active Perception Group](#), **Arizona State University**.
Advisor: [Dr. Yezhou Yang](#).
- Research Interests: Visual Question Answering, Multimodal Learning.
- 2018 **NLP Engineer.** Beijing, China. Jan–Jul 2018.
Search Engine Group. **Xiaomi**.
- Word segmentation, named entity recognition, unsupervised synonym pairs mining.
- 2016–17 **Machine Learning Engineer Full-Time & Intern.** Beijing, China. Oct 2016–Dec 2017.
Search Advertising Group. **Baidu**.
- Ads content generation, Query-Ads matching.
- 2016 **Machine Learning Engineer Intern.** Beijing, China. Jul–Sep 2016.
Growth Hacking Team. **Meituan**.
- Gradient boosted regression trees for new customer coupon distribution prediction.
- 2016 **Machine Learning Engineer Intern.** Beijing, China. Mar–Jun 2016.
Online Food Delivery Group. **Baidu**.
- Logistic regression model for customer CTR prediction.
- 2014–15 **Software Engineer Intern.** Beijing, China. Sep 2014–Feb 2015.
Beijing YunJiang Technology Co., Ltd.
- Web & mobile application backend development.

PROFESSIONAL MEMBERSHIPS

- 2022,26 Member, *Institute of Electrical and Electronics Engineers*.
- 2025– Member, *American Medical Informatics Association*.
- 2024–25 Associate Member, *Society of Critical Care Medicine*.
- 2024– Member, *Georgia Center for Diabetes Translation Research Data Repository*.
- 2023– Member, *Association for Computing Machinery*.
- 2024– Member, *Special Interest Group on Information Retrieval*.

SERVICES

Grant Reviews

- 2026 Reviewer Candidate. **NIH Early Career Reviewer** Program.
- 2025 Scientific Reviewer. **NIH AIM-AHEAD** Training Program.

Academic Services

- 2026 Area Chair. ACM Conference on Knowledge Discovery and Data Mining (**KDD**) – AI4Sciences Track.
- 2025 Scientific Program Committee Member. The 2026 AMIA Amplify Informatics Summit.
- 2025 Senior Program Committee Member. The 30th Pacific-Asia Conference on Knowledge Discovery and Data Mining (**PAKDD**).
- 2025 Mentor. IJCNLP-AAACL Student Research Workshop.
- 2025 Area Chair. *The 12th IEEE International Conference on Data Science and Advanced Analytics (DSAA)*.
- 2025 Poster Award Judge. *The 2025 Southeast Regional Clinical and Translational Science Conference*.
- 2022 Web Developer Student Lead. *28th ACM SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*.

Journal Peer Reviews

- 2026 npj Digital Medicine.
- 2023 Journal of Agricultural, Biological and Environmental Statistics (**JABES**).
- 2022–24 IEEE Transactions on Neural Networks and Learning Systems (**TNNLS**).
- 2021 IEEE Transactions on Knowledge and Data Engineering (**TKDE**).
- 2021 IEEE Transactions on Big Data (**TBD**).

Conference Peer Reviews

- 2022,25– ACM Conference on Knowledge Discovery and Data Mining (**KDD**).
- 2022,25– ACM Web Conference (**WWW**).
- 2023,25– AHLI Machine Learning for Health Symposium (**ML4H**)
- 2024– Association for Computational Linguistics Rolling Review (**ACL Rolling Review**).
- 2023 ACM Special Interest Group on Information Retrieval (**SIGIR**).
- 2021–22 Conference on Empirical Methods in Natural Language Processing (**EMNLP**).
- 2023 IEEE International Conference on Data Mining (**ICDM**).
- 2023 IEEE International Conference on Bioinformatics and Biomedicine (**BIBM**).
- 2022 AAAI Conference on Artificial Intelligence (**AAAI**).
- 2021 ACM International Conference on Information and Knowledge Management (**CIKM**).
- 2021 International Joint Conference on Natural Language Processing (**IJCNLP**).
- 2020 International Joint Conferences on Artificial Intelligence (**IJCAI**).
- 2019–23 IEEE International Conference on Robotics and Automation (**ICRA**).

Symposium & Workshop Peer Reviews

- 2024 AAAI Spring Symposium on Clinical Foundation Models.
- 2023 NeurIPS Workshop on Socially Responsible Language Modelling Research (**SoLaR**).
- 2023 ICLR Workshop on Trustworthy Machine Learning for Healthcare (**TML4H**).
- 2023–25 ACL SIGEDU Workshop on Innovative Use of NLP for Building Educational Applications (**BEA**).
- 2021–25 ICML Workshop on Interpretable Machine Learning in Healthcare (**IMLH**).
- 2021–24 ICCV Workshop on Computer Vision for Automated Medical Diagnosis (**CVAMD**).